

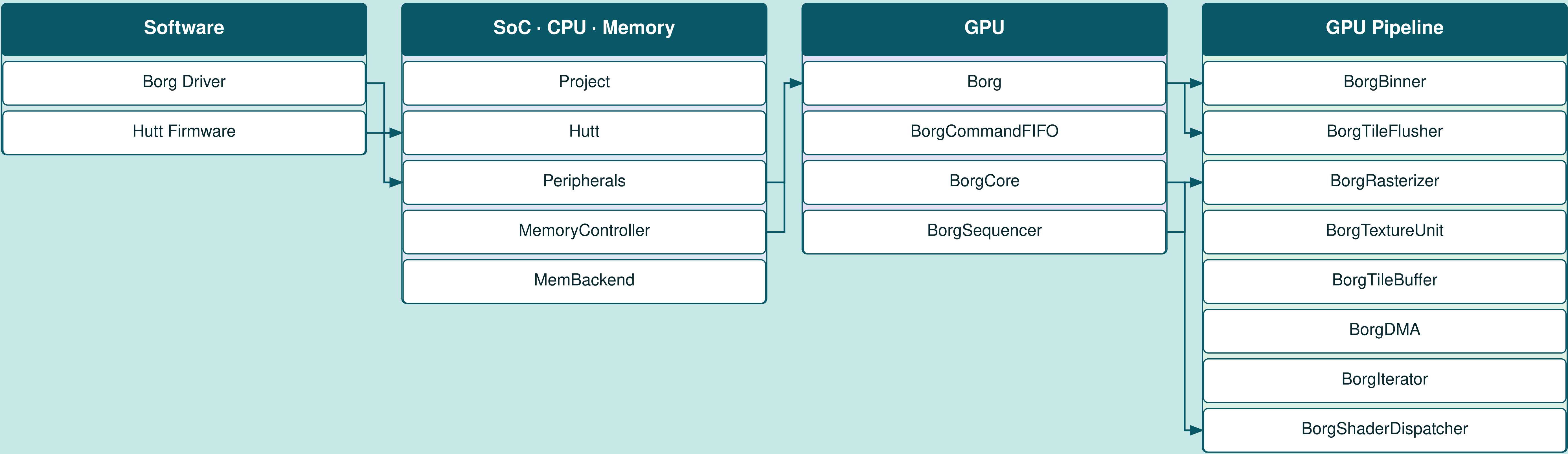
Borg (Bring yer Own GRaphics): An Open-Source Tile-Based GPU with Silicon Tapeout

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HPG 2026, July 17–19, Los Angeles, CA

SoC & TBDR Architecture

Borg is a complete **RISC-V SoC** featuring the **Hutt** core. The GPU follows the Tile-Based Deferred Rendering paradigm:



Geometry is binned tile-by-tile; shaded fragments are written to an on-chip tile buffer (4×4 pixels, FP16 RGBZ), minimising external memory bandwidth. The tile buffer is flushed to external memory only after all triangles covering that tile are rendered.

Motivation

- Every commercial GPU hides its microarchitecture behind NDAs
- This blocks **reproducibility**, **teaching**, and architectural experiments
- Concise:** GPU pipeline (**4.4k LoC** Chisel), Compiler (**1.6k LoC** Python), Firmware (**3k LoC** C).
- Open GPU simulators exist — but *no open GPU has reached silicon*

Borg is the fully open answer: a TBDR GPU **SoC** in Chisel, built with an open toolchain, and **submitted for ASIC tapeout**. Borg has contributed **75 patches upstream** to the open EDA ecosystem.

To our knowledge: the only open-source GPU with a submitted ASIC tapeout.

Development Roadmap: Phases 1–6

- Phase 1 ✓ — **Foundational Architecture** (2025)
- Phase 2 ✓ — **GPU Autonomy / 28-Day Sprint**
- Phase 3 — **Linux-Capable CPU** (M/A/MMU)
- Phase 4 — **Mesa Vulkan Driver** (NIR → SPIR-B)
- Phase 5 — **Hardware Extensions** (SIMD, Blending)
- Phase 6 — **Vulkan 1.0 Conformance**

Status Key Phase 2 Milestones

- ✓ 32-bit RISC-V ISA + Perspective Proj.
- ✓ Triangle Clipping + Fragment Interpolation
- ✓ Pixel Iterator (Rasterizer) + Z-Buffer
- ✓ Command FIFO + SystemRDL Register Map
- ✓ Texture Fetch + Shared Memory Controller
- ✓ GPU DMA Engine + Vertex Sequencer

Future Work & Open Questions

- 800×480 HDMI** native resolution (DDC/EDID)
- Higher fps via SDRAM burst writes and uniform caching
- SIMD-2 FMA for 2× fragment shader throughput
- BVH ray traversal ISA extensions (path tracing)

Open questions:

- Can a TBDR pipeline scale to multi-core tiles?
- Minimal ISA for hardware BVH traversal?
- How to best expose Borg via Mesa Vulkan?

The Transparency Gap: Borg vs. Commercial

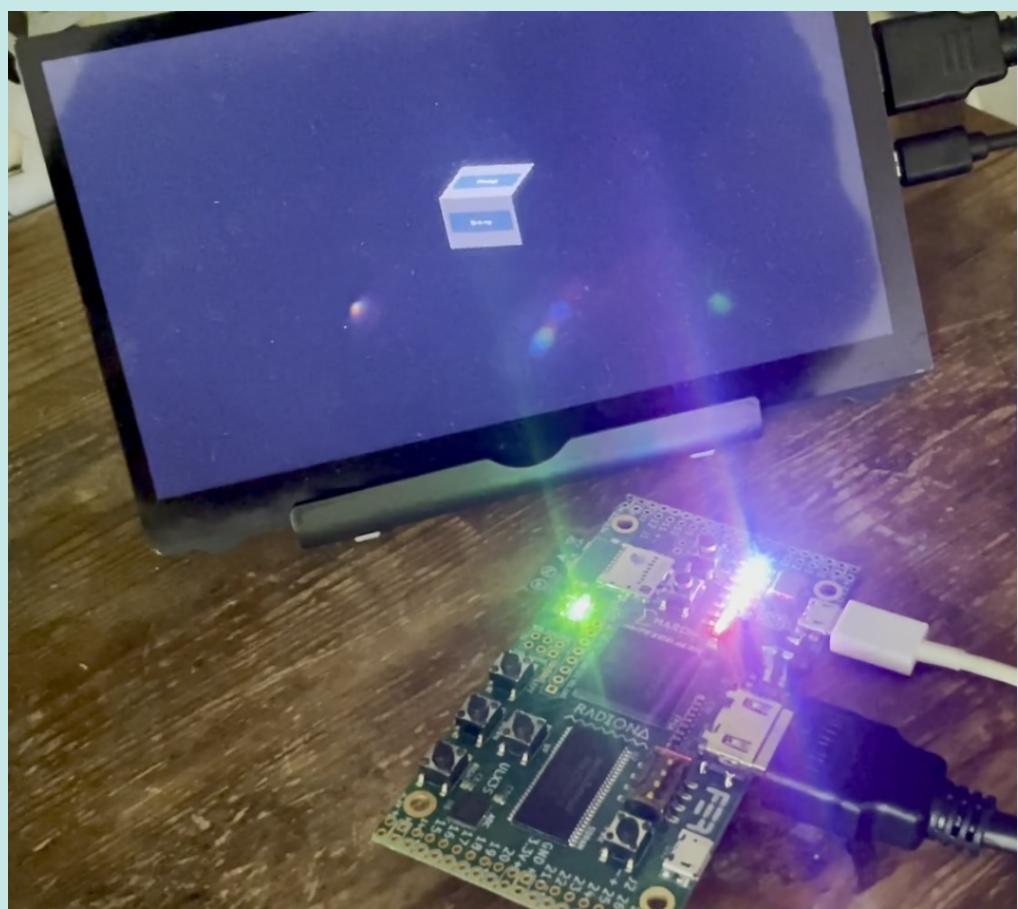
Feature	Borg	Mobile	Desktop
RTL Source	Open	Proprietary	Proprietary
EDA Toolchain	Libre	Proprietary	Proprietary
Silicon Layout	Public	Secret	Secret
ISA Specs	Public	Often NDA	Proprietary

Borg is the only design where researchers can inspect the layout and modify logic without NDAs.

Current Demo

Running on **ULX3S** (ECP5-85K) with **HDMI output**:

- 12 textured triangles, 128×128 framebuffer, double-buffered
- Full pipeline: vertex shader → texture fetch → Z-buffer → HDMI
- Orchestrated by integrated **Hutt** RISC-V core



VkCube running on ULX3S hardware with HDMI output

Open-Source Toolchain

HDL	Chisel (Scala, ~15k lines)
Synthesis	Yosys
P&R	nextpnr-ecp5 / LibreLane (Nix)
Simulation	Verilator, cocotb, Arcilator
Compiler	SPIR-V → SPIR-B (1.6k lines Python)
Firmware	32-bit RISC-V (~3k lines C)
Reg. map	PeakRDL-chisel
Math Cores	Berkeley Hardfloat (FMA)
PDKs	sky130A, gf180mcuD, ihp-sg13g2

Reproducible Build: direnv + Nix flakes.

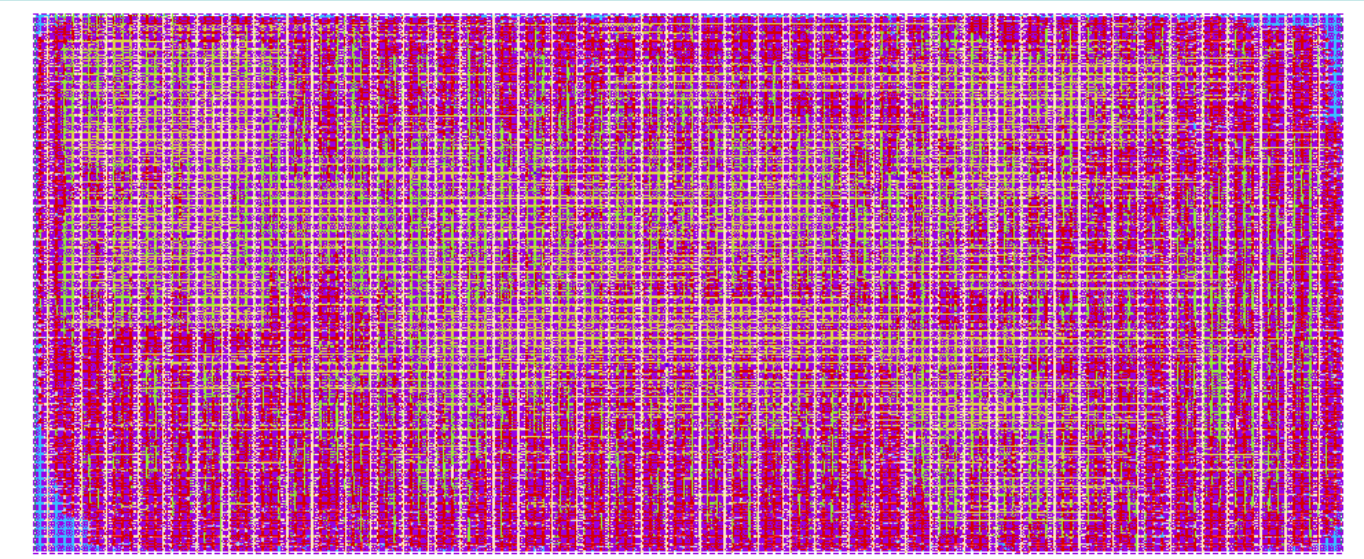
Zero-install setup for all 18+ EDA tools.

Ecosystem Impact: Pioneers the first fully reproducible ASIC flow in nixpkgs (LibreLane merged Jan 2026).

Comparison with Related Open GPUs

	Borg	Nyuzi	Vortex
Architecture	TBDR	GPGPU	RISC-V GPGPU
ISA	Custom	Custom	RV32
Tapeout	submitted	—	—
Open toolchain	✓	partial	partial
Target	ECP5	Arria 5	Xilinx

Physical Design (GDSII)



Metric	Value (TTIHP26a)
Technology	IHP SG13G2 (130 nm)
Die Area	1.72 × 0.71 mm ² (TT 8×4)
Std. Cells	107,525 instances
Clock / Slack	4 MHz / +75 ns setup, +0.1 ns hold
Power	2.33 mW @ 1.2 V, 25°C
DRC / LVS	✓ / ✓

ASIC Tapeouts (Tiny Tapeout)

Run	Process	Tiles	Content	Cost
TTIHP26a	IHP 130 nm	8×4	GPU SoC	515 €

Submitted; chips expected September 2026.

Funding



Borg is funded by the **NGI0 Commons Fund** (NLnet Foundation / European Commission, Grant No. 101135429) and Swiss SERI.

Design Values

Transparency No binary blobs, no hidden firmware.

Accessibility Runs on a \$100 FPGA; uses \$500 ASIC runs.

Simplicity Minimalist TBDR pipeline for maximum performance/area.

Selected References & Learn More

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Source Code



Book / Docs

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